Air Quality Prediction Using Machine Learning: Systematic Literature Review

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***Abstract*—Air pollution significantly impacts global health and the environment, causing millions of deaths annually. With rapid urbanization and industrialization, especially in countries like China, air quality has deteriorated due to pollutants such as PM2.5, PM10, CO_2, CO, NO_2, and SO_2. Traditional models often fail to predict air quality accurately due to their dependence on optimal parameter sets and data availability. Machine learning (ML) methods, however, offer superior performance by handling large datasets and complex variable relationships. This systematic literature review examines 77 articles from 2020 to 2024, highlighting the most effective ML methods for air quality prediction, including Random Forest Regression (RFR), Gradient Boosting Regression (GBR), Support Vector Machines (SVM), Long Short-Term Memory (LSTM) networks, and LightGBM. The review discusses the types of data used, evaluation metrics, and the models' performance. The findings emphasize the potential of ML to improve air quality monitoring and prediction, contributing to better public health and environmental management.**

Keywords—machine learning, prediction, air quality, systematic literature review

I. Introduction

Air pollution is a critical global issue, significantly affecting human health and the environment. According to the World Health Organization (WHO), air pollution is responsible for several million deaths annually, underscoring its severe impact on public health [1], [2]. The growing urbanization and industrialization, particularly in rapidly developing countries like China, have exacerbated air quality problems. In major cities, pollutants such as particulate matter (PM2.5, PM10), carbon dioxide (CO_2), carbon monoxide (CO), nitrogen oxides (NO_2), and sulfur dioxide (SO_2) are prevalent, significantly contributing to poor air quality [2], [3].

Machine learning (ML) has emerged as a powerful tool for predicting air quality due to its ability to handle large datasets and model complex relationships between variables [1], [4]. Traditional deterministic models, based on physical laws, often struggle with prediction accuracy due to their reliance on optimal parameter sets and data availability. In contrast, data-driven approaches using ML techniques, such as deep learning, ensemble learning, and broad learning methods, have demonstrated superior performance in predicting air quality[1], [2], [5].

Several studies have explored various ML methods for air quality prediction. For instance, Back Propagation Neural Networks (BPNN), Convolutional Neural Networks (CNN), Gated Recurrent Units (GRU), Long Short-Term Memory (LSTM), and Bi-directional LSTM (BiLSTM) have been employed to forecast Air Quality Index (AQI) with promising results [2], [4], [6]. These methods leverage historical AQI data and other relevant factors, such as meteorological conditions and pollutant concentrations, to provide accurate and timely air quality forecasts [2], [4].

In China, extensive research has been conducted using ML models to predict AQI across different cities. For example, studies have shown that BiLSTM outperforms other approaches in short-term AQI prediction, particularly when using historical data from multiple observation sites [2], [7], [8]. Similarly, research in Hangzhou has demonstrated the effectiveness of combining seasonal factors with ML models like Sparrow Search Algorithm Extreme Learning Machine (SSA-ELM) to predict PM2.5 concentrations accurately [7].

Despite these advancements, several challenges remain in the field of air quality prediction using ML. These include the variability and quality of data, the need for robust preprocessing techniques, and the integration of spatial and temporal factors into predictive models [2], [4]. Addressing these challenges requires a comprehensive understanding of the existing literature and the identification of gaps and opportunities for future research [4], [9].

This systematic literature review aims to provide a thorough examination of the current state of air quality prediction using ML. By reviewing approximately 77 articles, this study will explore the various ML methods used, the types of data and preprocessing techniques employed, the evaluation metrics and results reported, and the challenges and future directions in the field [4]. Through this review, we aim to synthesize existing knowledge, identify best practices, and suggest avenues for further research to enhance the accuracy and applicability of ML models in air quality prediction.

Further, integrating meteorological data into ML models enhances the precision of air quality forecasts by capturing the influence of weather conditions on pollutant concentrations. Studies using Graph Convolutional Networks (GCN) and Exogenous Long Short-Term Memory (E-LSTM) networks have shown promising results in modeling the spatiotemporal dynamics of pollutants like PM2.5, demonstrating the

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superiority of these advanced techniques over traditional models [1], [7]. These innovative approaches underscore the potential of ML to transform air quality prediction and contribute significantly to public health and environmental management.

II. Methods

This study aims to identify and analyze the machine learning methods used for air quality prediction, the types of data used, and the evaluation and performance of these models. This Systematic Literature Review (SLR) will review 77 articles from various relevant sources.

A. Search Strategy

The search strategy involves several steps, which are illustrated in the following figure. First, the databases IEEE, Science Direct, and ACM were used. The keyword combinations included "Machine Learning," "Prediction," and "Air Quality." The process started with identifying the main keywords and then using various combinations of these keywords to search the databases. Results were filtered based on relevance and publication period (2020-2024). Articles that met the criteria were collected, and their full texts were downloaded for further analysis. The search strategy steps are depicted in the following Fig. 1. In Fig. 2, we show the distribution of the final papers according to the search engines from which they were retrieved.

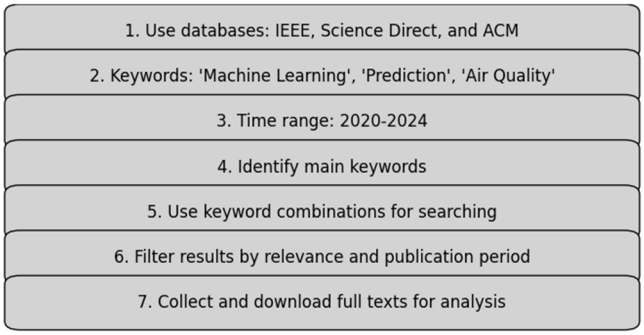


Fig. 1. Search stategy steps

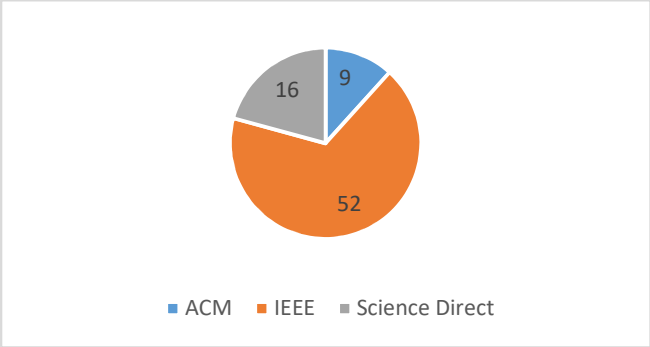


Fig. 2. Selected papers by database

B. Research Question

To guide and focus the review, the following research questions will be used:

RQ1 : Why is air quality prediction important and how does this research contribute to that understanding?

RQ2 : What types and sources of data are used to train the machine learning models?

RQ3 : What methods are most commonly used and how do they perform compared to each other?

RQ4 : What evaluation methods are used and how is the performance of these models measured and compared?

III. Result and Discussion

A. Main Objectives

Air quality prediction is crucial due to its significant impact on public health and the environment. Poor air quality is linked to numerous health problems, including respiratory and cardiovascular diseases, and can lead to substantial social and economic costs. Accurate air quality predictions enable timely interventions and inform public health advisories, mitigating adverse health effects and improving quality of life. This research contributes to understanding air quality prediction by systematically reviewing recent advancements in machine learning techniques. It highlights the effectiveness of these methods in handling large datasets and modeling complex relationships between pollutants and meteorological factors, thus enhancing prediction accuracy and providing valuable insights for policymakers and researchers [10], [11], [12], [13].

B. Data Used

The data used to train machine learning models for air quality prediction typically include pollutant concentration levels (such as PM2.5, PM10, NO_2, SO_2, CO, and O_3), meteorological data (temperature, humidity, wind speed, and direction), and historical air quality index (AQI) values. Sources of this data include government environmental monitoring agencies, satellite remote sensing data, and ground-based sensor networks. For example, air quality data for Henan Province was sourced from the National Earth System Science Data Center, while other studies utilized data from the U.S. Geological Survey's Earth Explorer database and local environmental protection departments [5], [10], [11], [13].

C. Machine Learning Methods

The most commonly used machine learning methods for air quality prediction include Random Forest Regression (RFR), Gradient Boosting Regression (GBR), Support Vector Machines (SVM), and various neural network architectures such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and K-Nearest Neighbors (KNN). Ensemble methods like RFR and GBR often outperform other algorithms in terms of accuracy and generalization ability. For instance, RFR and GBR demonstrated superior performance in predicting the Air Quality Index (AQI) and pollutant concentrations, achieving lower Mean Squared Error (MSE) and higher R^2 values compared to linear regression and simpler models. Additionally, advanced methods such as XGBoost and CatBoost have shown high accuracy and robustness in several studies, proving their effectiveness in various environments.

TABLE I. Overview of Methods Used in Reviewed Papers

Research	Method Used
[14], [15], [16], [17], [18]	Random Forest Regression (RFR)
[3], [19], [20], [21]	Gradient Boosting Regression (GBR)
[22], [23]	Support Vector Machine (SVM)
[19], [24], [25], [26]	Long Short-Term Memory (LSTM)
[12], [13], [27], [28]	Convolutional Neural Network (CNN)

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[7], [29]	Deep Neural Network (DNN)
[30]	XGBoost
[31]	CatBoost
[32]	K-Nearest Neighbors (KNN)
[33]	Elman Neural Network with WOA
[33]	Improved Whale Optimization Algorithm (IWOA)
[19], [34]	Recurrent Neural Network (RNN)
[31], [35]	LightGBM
[36]	Deep Belief Network-Extreme Learning Machine (DBN-ELM)
[37]	ARIMA Time Series Model
[38]	Neural Network Implementations
[39]	Ensemble Models and Time Series Decomposition
[3]	Gradient Boosted Trees, Neural Network
[31]	Wavelet Decomposition and BPNN
[40]	Stacking Ensemble Model

Below are the results of the review of the most popular methods as shown in Table 1.

1) Random Forest Regression (RFR): Is an ensemble learning method that builds multiple decision trees during training and outputs the mean prediction of the individual trees. RFR is particularly effective for air quality prediction due to its ability to handle large datasets and complex interactions among variables. It has demonstrated strong performance in predicting pollutant concentrations and AQI, often achieving lower Mean Squared Error (MSE) and higher R^2 values compared to simpler models like linear regression [20], [24], [41].

2) Gradient Boosting Regression (GBR): Is another powerful ensemble learning technique that builds trees sequentially, with each new tree correcting errors made by the previous ones. GBR is known for its high accuracy and robustness in air quality prediction tasks. It has been used successfully to predict AQI and pollutant levels, often outperforming other algorithms in terms of prediction accuracy and generalization ability [20], [21], [42].

3) Support Vector Machine (SVM): Is a supervised machine learning model widely used for classification and regression tasks. In air quality prediction, SVM transforms linearly separable data into high-dimensional space using kernel functions, such as the Radial Basis Function (RBF). This allows for better handling of non-linear relationships between features. SVM models with RBF kernel have been shown to outperform other models in predicting the Air Quality Index (AQI) by providing superior results on various assessment metrics like R^2 score and Root Mean Squared Error (RMSE) [10], [31], [39].

4) Long Short-Term Memory (LSTM): Networks are a type of recurrent neural network (RNN) designed to better capture long-term dependencies in sequential data. LSTMs are particularly effective in modeling time-series data and have shown state-of-the-art results in air quality prediction. They can handle the non-linearity and instability in pollutant concentration data, making them suitable for predicting AQI over various time horizons. Studies have demonstrated the effectiveness of LSTMs in providing accurate and stable predictions of PM2.5 and other pollutants [2], [6], [10], [22], [29], [39].

5) LightGBM: Is a gradient boosting framework that uses tree-based learning algorithms. It is known for its efficiency and speed in training large datasets. LightGBM has demonstrated exceptional performance in air quality prediction tasks, achieving high accuracy and robustness. For instance, a LightGBM model achieved an accuracy of 99.75% in classifying air quality levels, making it one of the most effective methods for this purpose [10], [12], [39].

6) Various neural network implementations: Including standard feedforward neural networks and more complex architectures, have been used in air quality prediction. While these models can capture complex relationships in data, they sometimes achieve lower accuracies due to challenges in tuning and variability in data. Some neural network models have reported accuracies as low as 74% in certain cases, highlighting the need for careful model selection and optimization [10], [12], [39].

D. Model Evaluation and Metrics

The performance of air quality prediction models is typically evaluated using metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics measure the accuracy of the predicted values compared to the actual observed values. Cross-validation techniques, such as ten-fold cross-validation, are also used to ensure the models' generalization capabilities. For example, a study comparing various regression algorithms found that RFR and GBR achieved the lowest MSE and MAE, indicating high prediction accuracy and robustness. Ensemble methods like XGBoost and CatBoost also performed exceptionally well, often surpassing traditional models in accuracy and reliability [5], [20], [31].

Here are the algorithms and their performance evaluation results.

1) Random Forest Regression (RFR): RFR models have shown strong performance in predicting pollutant concentrations and AQI. For example, an RFR model used to predict air quality in the Wuling mountain area achieved an MSE of 3.54 and an MAE of 1.28, demonstrating its effectiveness compared to other models like decision trees and deep back propagation neural networks [18], [20].

2) Gradient Boosting Regression (GBR): GBR models also performed well, often achieving lower MSE and MAE compared to other models. For instance, a study found that GBR had an MSE of 5.85 and an MAE of 5.16 when used for air quality prediction in Henan Province, outperforming models like Support Vector Regression (SVR) and LSTM in terms of RMSE and MAE [20], [43].

3) Support Vector Machine (SVM): SVM models, particularly with RBF kernels, have been shown to outperform other models in predicting AQI. A study reported that SVM with RBF kernel achieved an R^2 of 0.9989, an MSE of 24.44, and an RMSE of 4.94, highlighting its superior performance in AQI prediction compared to linear and polynomial kernels [39].

4) Long Short-Term Memory (LSTM): LSTM networks have demonstrated state-of-the-art results in time-series air quality prediction tasks. For example, an LSTM model for AQI prediction in Ahmedabad city achieved an R^2 of 0.9512, an MSE of 656.62, and an RMSE of 25.62. This shows the

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model's effectiveness in capturing long-term dependencies in the data [39].

5) LightGBM: LightGBM, a gradient boosting framework, has been noted for its high accuracy in air quality prediction tasks. For instance, a LightGBM model achieved an accuracy of 99.75% in classifying air quality levels, making it one of the most effective methods for this purpose. Its ability to handle large datasets efficiently and its fast training speed contribute to its superior performance [10], [12], [39].

6) Neural Network Implementations: Certain neural network implementations have been observed to have lower accuracies in some cases, achieving accuracies as low as 74%. This variability can be attributed to the complexity of the data and the challenges in tuning neural network models effectively [10], [12], [39].

IV. Conclusion

This systematic literature review highlights significant advancements and challenges in using machine learning (ML) techniques for air quality prediction. Reviewing 77 articles from 2020 to 2024, it demonstrates that methods like Random Forest Regression (RFR), Gradient Boosting Regression (GBR), Support Vector Machines (SVM), Long Short-Term Memory (LSTM) networks, and LightGBM outperform traditional models by effectively handling large datasets and complex variable relationships. RFR and GBR exhibit high accuracy and robustness, while SVM with RBF kernels excels in predicting the Air Quality Index (AQI). LSTM networks accurately model time-series data, and LightGBM achieves remarkable accuracy in classifying air quality levels. Data typically includes pollutant concentrations, meteorological factors, and historical AQI values, sourced from environmental monitoring agencies, satellite data, and sensor networks. Model performance is evaluated using metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2, with ensemble methods like XGBoost and CatBoost often surpassing traditional models. Despite advancements, challenges such as data variability, preprocessing needs, and integrating spatial and temporal factors persist. Future research should focus on incorporating meteorological data, optimizing advanced techniques like Graph Convolutional Networks (GCN) and Exogenous LSTM (E-LSTM), and improving neural network models to enhance air quality prediction accuracy, thereby contributing to better public health and environmental management.

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